FET physics

Cheat sheet · Day 15 · Sat 7 Nov · pack E9

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
FET PHYSICS (Week 6)                          I_G = 0  ->  I_S = I_D
JFET / D-MOSFET (Shockley, saturation):
  I_D = I_DSS (1 - VGS/VP)^2       VGS = VP (1 - sqrt(I_D/I_DSS))
  I_DSS: VGS = 0, VDS >= |VP|  (max for JFET; D-MOSFET can exceed with VGS>0)
  JFET: I_D = 0 for VGS beyond VP (VP = VGS(off))
  Sketch: (0, I_DSS) (0.3VP, I_DSS/2) (VP/2, I_DSS/4) (VP, 0)  concave up
  Ohmic: r_d = r_o / (1 - VGS/VP)^2     Knee: VDS(sat) = VGS - VP
E-MOSFET:
  I_D = k (VGS - VT)^2  for VGS > VT (n), else 0      no I_DSS!
  k = I_D(on) / (VGS(on) - VGS(Th))^2   [mA/V^2]
  VGS = VT +/- sqrt(I_D/k): keep the root that turns it ON
  VDS(sat) = VGS - VT
p-CHANNEL: same equations, signed values.
  p-JFET, p-DMOS: VP > 0.  p-EMOS: VT < 0, on for VGS < VT.  VDS < 0.
OUTPUT char = I_D vs VDS (VGS steps).  TRANSFER = I_D vs VGS (in saturation).
  Output -> transfer: read I_D at each VGS at one saturation VDS.
INVERTER / CMOS:
  NM_L = V_IL - V_OL     NM_H = V_OH - V_IH
  Ideal: V_OH = VDD, V_OL = 0, V_IL = V_IH = VDD/2 -> NM = VDD/2
  CMOS: VGS,n = Vi ; VGS,p = Vi - VDD.  Vi=0 -> pMOS on, Vo=VDD; Vi=VDD -> nMOS on, V
  One device always off -> ~no static power
  t_P = (t_PLH + t_PHL)/2, each measured to the output crossing VDD/2
```